Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_modular_operator_diversity_residual_replay.
- Best held-out TSPLIB gap: phase7_full_solver_evolution.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260520_071708_q
- started_at_local: 2026-05-20 07:17:08
- finished_at_local: 2026-05-20 07:51:47
- duration_hhmm: 00:35
- duration_seconds: 2079.291
- seed_offset: 16000
- replicate_label: q
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.213387	0.064916	0.159398	0.802078	0.56	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.236585	0.064041	0.12761	0.887686	0.4475	False	0.003318	0.00507
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235058	0.054968	0.121317	0.828977	0.435	False	0.004105	0.00874
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.235058	0.049737	0.118013	0.819295	0.4425	False	0.00861	0.892423

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235058	0.050827	0.118702	0.889017	0.44	False	0.00784	0.911918
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.235667	0.054968	0.121542	0.0	0.5175	False	0.006779	0.272308

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.213387, family holdout 0.064916, combined 0.159398.
- Accepted novelty 0.802078, complexity 0.56, and adaptation efficiency -0.031051.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.236585, family holdout 0.064041, combined 0.12761.
- Accepted novelty 0.887686, complexity 0.4475, and adaptation efficiency 0.002106.
- Last-epoch runtime 76.976775 ms and distance evaluations 5024.25.
- Validation: surviving False, transplant delta 0.003318, positive scaffolds 0, Pareto runtime inflation 0.00507.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 0.003718, "gap_delta": 0.006293, "runtime_inflation": 0.00507, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.828977, complexity 0.435, and adaptation efficiency 0.004785.
- Last-epoch runtime 79.784925 ms and distance evaluations 5142.25.

- Validation: surviving False, transplant delta 0.004105, positive scaffolds 0, Pareto runtime inflation 0.00874.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": 0.00874, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.049737, combined 0.118013.
- Accepted novelty 0.819295, complexity 0.4425, and adaptation efficiency 0.009279.
- Last-epoch runtime 74.78395 ms and distance evaluations 4683.0.
- Validation: surviving False, transplant delta 0.00861, positive scaffolds 1, Pareto runtime inflation 0.892423.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.884148, "gap_delta": -0.005462, "runtime_inflation": 0.892423, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.050827, combined 0.118702.
- Accepted novelty 0.889017, complexity 0.44, and adaptation efficiency 0.031794.
- Last-epoch runtime 89.91765 ms and distance evaluations 4278.75.
- Validation: surviving False, transplant delta 0.00784, positive scaffolds 1, Pareto runtime inflation 0.911918.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.929896, "gap_delta": -0.003304, "runtime_inflation": 0.911918, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.0, complexity 0.5175, and adaptation efficiency 0.0.
- Last-epoch runtime 150.94045 ms and distance evaluations 7315.75.
- Validation: surviving False, transplant delta 0.006779, positive scaffolds 1, Pareto runtime inflation 0.272308.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 16, "complexity_score": 0.52, "distance_eval_inflation": 0.19862, "gap_delta": 0.000224, "runtime_inflation": 0.272308, "same_gap_faster": false}

Judge Appendix

Summary of TSP Modular-Operator Discovery Results

Main Transfer Evidence (Held-Out TSPLIB & Family Gaps)

Condition	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Transplant Mean Gap Delta	Family Signal (Mean Gap Delta)	Surviving Candidate?
Best TSPLIB: phase7_full_solver_evolution	0.213387	0.064916	0.159398	0.0	N/A	No

Condition	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Transplant Mean Gap Delta	Family Signal (Mean Gap Delta)	Surviving Candidate?
Best Transfer: phase7_modular_operator_diversity_residual_replay	0.235058	0.049737	0.118013	0.00861	-0.004484 (improvement in some families)	No
phase7_modular_operator_evolution	0.236585	0.064041	0.12761	0.003318	+0.007995 (positive but no improvement families)	No
phase7_modular_operator_random_replay	0.235058	0.054968	0.121317	0.004105	0.0	No
phase7_modular_operator_compression_pressure	0.235058	0.050827	0.118702	0.00784	-0.00355 (small improvement in 1 family)	No
phase7_modular_operator_pareto_selection	0.235667	0.054968	0.121542	0.006779	+8.7e-05 (negligible)	No
phase7_baseline_heuristic_only	0.235058	0.054968	0.121317	0.0	N/A	No

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Operator Validation, Transplant, Ablation, and Pareto Tradeoffs Highlights

- **No operator survived transplant or ablation checks:**
- All modular operators fail to show a significant, consistent positive impact when disabled (ablation results show no significant degradation; disabling operators often unchanged or better gap).
- No operators are surviving candidates.

- **Transplant results:**
- None of the operators improve significantly when transplanted into other scaffolds (no positive scaffold counts except rare cases with negligible or negative family signals).
- Several operators fail to transplant cleanly into alternative solvers (notably clustered_local_search, sparse_three_opt).

- **Family Holdout Performance:**
- No operator shows consistent improvement over host baselines across validation families. Some show regressions on grid_like_tsp and no positive family gains in heldout TSPLIB.
- The restart controllers show some positive gains on "clustered_tsp" and "two_cluster_bottleneck_tsp" families but at high runtime inflation (near 90% increase).

- **Pareto tradeoffs:**
- Runtime inflation is generally non-negligible (up to ~0.9) often without corresponding gap improvements.
- No operator demonstrates a clear better-gap-at-same-runtime scenario.

- **Core Ideas and Novelty:**
- Operators implement interesting ideas: deterministic local ranking for move selection, stagnation escape via perturbations, adaptive restart control, and candidate pruning.
- However, these do not materialize into reusable, validated operators due to failure in ablation and transplant checks.

Conservative Interpretation and Conclusions

- **Main Known Result:** The "phase7_full_solver_evolution" condition (non-modular) achieves the best TSPLIB gaps but offers no evidence of reusable modular operator structure.

- **Modular Operators:**
- None survive ablation or transplant validations meaning no modular operators have demonstrated generalizable, reusable impact on held-out or family holdout test sets.

- **Transfer Evidence:**
- Although some modular operator variants show small improvements in certain families (e.g., restart controllers on clustered/bottleneck instances), these improvements come at a significant runtime cost and fail ablation tests, undermining claims of operator effect.

- **Pareto Tradeoffs:**
- No operator distinctly improves Pareto frontier (better gap without runtime cost or faster runtime at same gap).

- **No Algorithm Discovery:**
- Since none survive scaffold or family transfer tests, no claim can be made about discovered modular algorithms or reusable operators.

Recommendation

- **No modular operator can currently be deemed interesting or reusable** based on the given evidence.

- Future work should focus on improving operator validation rigor, considering operator impact on transfer families, ensuring positive ablation signals, successful transplants, and better runtime-gap tradeoffs before claiming discovery of reusable modular operators.

Summary Table of Modular Operators (Key Validation Metrics)

Operator Name	Type	Final TSPLIB Gap	Family Gap	Transfer Gap	Ablation Effect	Transplant Effect	Runtime Inflation	Surviving Candidate?
candidate_ranker_bottleneck_aware_2opt_moves	candidate_ranker	0.236585	0.064041	0.12761	Disabling helps	No positive scaffolds	~0.5%	No
perturb_escape_double_bridge_progressive	perturbation	0.235058	0.054968	0.121317	No effect	No positive scaffolds	~0.87%	No
restart_controller_deterministic_perturb_restart_pool	restart_controller	0.235058	0.049737	0.118013	Disabling helps	1 positive scaffold (marginal)	~89%	No

Operator Name	Type	Final TSPLIB Gap	Family Gap	Transfer Gap	Ablation Effect	Transplant Effect	Runtime Inflation	Surviving Candidate?
descriptor_driven_restart_controller_stagnation_escape	restart_controller	0.235058	0.050827	0.118702	Disabling helps	1 positive scaffold (marginal)	~91%	No
structure_aware_pruner_midlimit	candidate_pruner	0.235667	0.054968	0.121542	Disabling helps	1 positive scaffold (marginal)	~27%	No

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Final conservative conclusion:

No modular operator in the suite can be claimed as reusable or algorithmically discovered under the current criteria focusing on held-out transfer gap, transplant, ablation validation, and Pareto tradeoffs.